

ST HELENS, Australia

WMO: 959810

Lat: 41.3381S

Lon: 148.2792E

Elev: 161

StdP: 14.61

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.2	36.9	27.9	22.0	46.3	29.9	24.1	46.4	22.6	52.7	20.5	53.2	7.4	320	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	79.8	64.1	75.6	63.4	72.8	62.6	68.0	73.8	66.5	71.6	65.2	70.2	11.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.0	70.0	64.8	92.3	68.7	63.3	87.6	67.6	32.4	73.5	31.3	72.0	30.2	70.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.3	16.1	DB	32.1	91.3	0.8	4.3	31.5	94.4	31.0	97.0	30.6	99.4	30.0	102.6	(4)
(5)			WB	31.1	70.6	0.9	1.6	30.5	71.8	30.0	72.7	29.5	73.6	28.8	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	63.6	62.9	61.3	57.0	52.8	49.5	48.4	49.1	51.4	54.3	57.8	60.6
	DBStd	6.82	4.93	4.19	4.80	4.36	4.10	3.64	3.38	3.69	3.81	4.56	4.83	4.49
	HDD50	246	0	0	0	2	17	53	74	63	27	9	1	0
	HDD65	3521	85	84	137	243	380	466	516	494	407	334	224	151
	CDD50	2319	422	362	350	212	103	37	23	34	70	142	234	330
	CDD65	114	41	25	22	2	0	0	0	0	0	2	7	15
	CDH74	562	229	86	65	8	0	0	0	0	0	14	70	90
	CDH80	148	78	20	11	0	0	0	0	0	0	1	17	21
Wind	WSAvg	8.1	8.9	7.8	7.5	7.5	7.6	7.6	7.6	8.0	8.0	8.5	8.7	8.9
Precipitation	PrecAvg	29.4	2.4	1.7	2.4	2.6	2.5	2.2	2.8	2.6	2.4	2.3	2.7	3.4
	PrecMax	44.7	10.6	7.3	7.0	9.8	7.8	6.6	6.2	7.8	7.2	6.7	7.4	15.4
	PrecMin	18.0	0.2	0.0	0.5	0.1	0.6	0.6	0.2	0.0	0.4	0.1	0.1	0.2
	PrecStd	6.9	2.2	1.6	1.6	2.3	1.8	1.3	1.7	1.8	1.6	1.5	1.8	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	83.4	80.2	74.4	66.7	61.8	60.0	63.0	67.9	75.9	82.5	83.0
		MCWB	66.9	66.3	65.4	61.9	57.5	54.6	52.5	53.1	55.1	57.9	64.0	64.8
	2%	DB	79.8	75.7	74.9	69.9	63.9	59.7	58.0	60.2	64.1	69.2	73.8	75.6
		MCWB	63.9	64.9	64.9	60.4	56.4	53.8	51.3	51.9	53.9	56.6	60.2	62.3
	5%	DB	74.8	72.8	71.6	66.9	62.1	58.2	56.5	58.0	61.3	65.4	69.3	71.8
		MCWB	63.0	64.1	63.4	59.5	55.8	52.8	50.6	51.0	52.7	55.5	59.0	61.3
	10%	DB	71.7	70.3	69.0	64.6	60.6	56.6	55.1	56.3	59.0	62.6	66.2	68.8
		MCWB	63.0	63.1	62.3	58.6	54.9	51.6	49.8	50.3	51.8	54.4	58.1	59.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.2	69.4	69.1	64.7	60.4	57.6	54.4	55.4	57.9	61.1	65.7	67.3
		MCDB	78.5	76.0	74.3	69.1	62.5	59.5	57.1	58.2	62.3	65.6	73.5	75.5
	2%	WB	67.6	67.8	67.2	62.9	58.5	55.1	52.9	53.8	56.0	59.2	63.3	65.2
		MCDB	74.8	72.3	71.8	66.8	62.2	57.9	56.0	57.7	61.1	64.0	70.7	71.2
	5%	WB	65.9	66.4	65.5	61.3	57.0	53.5	51.8	52.3	54.3	57.6	61.2	63.5
		MCDB	71.2	70.4	69.9	65.5	61.4	56.7	55.3	56.8	59.5	63.1	66.9	68.8
	10%	WB	64.4	64.8	63.6	59.4	55.4	52.0	50.6	51.0	52.9	55.9	59.3	61.6
		MCDB	70.0	69.0	68.2	64.0	59.8	55.7	54.5	55.7	58.0	62.1	64.4	66.9
Mean Daily Temperature Range	5% DB	MDBR	14.7	13.0	13.8	13.9	13.7	12.9	13.1	14.2	15.0	14.8	14.7	14.5
		MCDBR	21.1	17.2	17.5	17.0	14.6	14.1	13.6	15.5	18.2	18.8	20.6	19.6
		MCWBR	9.5	9.1	9.4	9.6	9.4	9.3	9.0	9.5	10.5	10.1	10.0	9.7
	5% WB	MCDBR	17.2	13.5	14.7	14.4	12.3	12.0	11.8	13.9	15.1	15.6	17.0	15.9
		MCWBR	9.3	8.8	8.8	9.3	8.9	8.5	8.7	9.3	9.8	9.8	9.6	9.3
Clear-Sky Solar Irradiance	taub	0.351	0.346	0.336	0.321	0.305	0.296	0.294	0.310	0.327	0.338	0.345	0.344	
	taud	2.516	2.546	2.567	2.586	2.599	2.610	2.628	2.576	2.524	2.500	2.510	2.522	
	Ebn at Noon	310	303	292	275	259	251	261	275	290	302	310	314	
	Edh at Noon	35	32	29	24	20	18	19	24	30	33	35	35	
All-Sky Solar Radiation	RadAvg	2110	1776	1422	972	653	515	585	851	1258	1649	1919	2167	
	RadStd	142	95	68	55	36	40	26	59	89	80	132	131	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+0.59	N/A	N/A	+1.24	+0.61	N/A	N/A	-184	+196	+32

Nomenclature: See separate page